



Shane Rowand

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Mechanical Engineering student seeking a Mechanical/Robotics Engineering internship

Education

Rensselaer Polytechnic Institute

Troy, NY

B.S. in Mechanical Engineering — Expected May 2030

- Relevant Coursework: Introduction to Robotics, Computer-Aided Design (Siemens NX), Multivariable Calculus, Linear Algebra, Physics I & II, Computer Science I

Technical Skills

Mechanical Design: Siemens NX, SolidWorks, Fusion 360, OnShape

Fabrication & Manufacturing: 3D printing/prototyping, Tinkercad, woodworking, resin casting, basic electrical wiring

Robotics & Controls: MuJoCo/MJX, reinforcement learning (PPO), URDF, Docker

Programming: Python, self-taught software development

Platforms: Windows, Git, VS Code

Projects

Axle — Legged-Robot Platform & Personal AI Assistant

Personal Project

- Extending an AI agent platform into a hexapod legged robot, architecting a two-layer control system that separates high-level LLM reasoning from a low-level locomotion policy
- Training a reinforcement learning locomotion policy (PPO, domain randomization) in MuJoCo/MJX at 50–200 Hz control rates, developing a single-leg prototype before scaling to the full hexapod
- Modeling leg assemblies in CAD and exporting URDF models to integrate mechanical design directly into the simulation pipeline
- Built the underlying agent core (Python, WebSocket architecture, SQLite-backed session memory) that will serve as the robot's high-level reasoning layer
- Implemented Docker-sandboxed execution and multi-modal desktop control (computer-use automation, browser bridge, voice input) to prototype safe agent-hardware interaction patterns

Seatbelt Usage Tracker — PennDOT Competition

Prototype

- Engineered a clamp-on mechanical device to track seatbelt buckle usage, addressing a road-safety data problem posed by PennDOT
- Modeled the housing in Tinkercad and 3D printed a working prototype, iterating on fit and function to present at the competition

Spotify Barcode Sign

Personal Project

- Fabricated a 3-foot wooden sign in walnut and cherry, precisely machining dimensions to match a scannable Spotify barcode linking to a custom playlist
- Cast resin inlays and wired an embedded LED strip around the border, combining woodworking, casting, and basic electrical design

Activities & Leadership

RPI Robotics Club

Troy, NY

Member

2026 – Present

- **Combat Robot:** Designed a 11lb combat robot in OnShape; 3D printed and manufactured all components, then assembled, wired, and tested it in competition against other teams

Work Experience

Weis Markets

Perkasie, PA

Produce Associate

2024 – Present